The Venkatesh Lab
BRIGHAM AND WOMEN'S HOSPITAL
HARVARD MEDICAL SCHOOL
Understanding the Neuroscience of Cancers

HARVARD UNIVERSITY

Characterization of the Effects of Tumor–Brain Communication on Neural Activity and Region-Specific Behavior in Small Cell Lung Cancer

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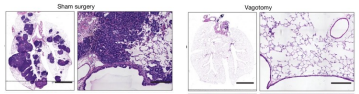
Introduction

Small Cell Lung Cancer (SCLC)

- SCLC is a rare, fast-growing cancer and the most fatal subtype of lung cancer, accounting for 15% of cases.
- SCLC cells derive from pulmonary neuroendocrine cells and have been shown to have neuronal features (Hartmann & Sage, 2024).
- Many patients also exhibit metastases to the brain.
- This indicates a need to explore interactions between SCLC and the nervous system.

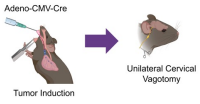
Vagus Nerve Innervation

Previous work by the lab has demonstrated the significance of the vagus nerve in the growth of SCLC tumors:



Savchuk et al., 2025

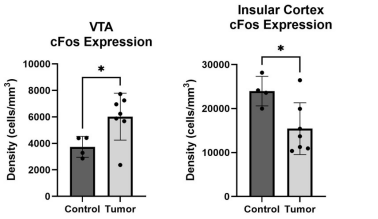
Adeno-CMV-Cre intratracheal injection into triple knockout (TKO) SCLC mouse models causes the deletion of three tumor suppressor genes and induces tumor formation (Savchuk et al., 2025, Schaffer et al., 2010).



Unilateral cervical vagotomies can be used to reduce tumor burden in TKO SCLC mouse models.

Changes in Brain Activity

- Whole brain clearing showed changes in brain activity of male tumor-burdened mice in the **ventral tegmental area** and **insular cortex**.



Graphs by Alexis Franklin

AIM #1: Confirm tumor-induced changes in neural activity in the **ventral tegmental area (VTA)** and the **insular cortex (IC)**

AIM #2: Determine behavioral changes linked specifically to the **VTA** and **IC** regions of the brain

Methodology

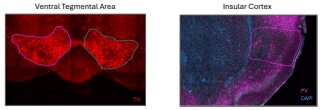
AIM #1: Histology

Using an early tumor/control cohort of male TKO mice,

1. Sectioned the entire brain into 50-micron coronal slices

2. Stained slices with cFos, tyrosine hydroxylase (TH), and parvalbumin (PV)

cFos → marker for (recently) active neurons
TH → marker for dopaminergic neurons, used to locate the VTA
PV → marker for a subtype of inhibitory (GABAergic) interneurons, used to locate the IC



3. Analyzed cFos cell density using cell counting software

AIM #2: Behavior

Using an early vagotomy/sham vagotomy cohort of male TKO mice,

Tested for changes in behavior linked to both the VTA and the IC: **anxiety, reward behavior**, and **social interaction**

Open Field Test



Operant Conditioning



Light/Dark Box

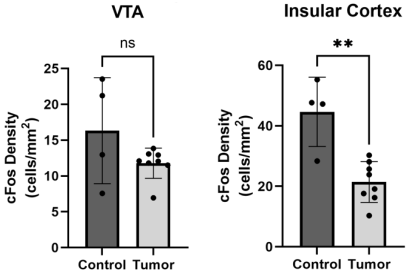


Three Chamber Social Test

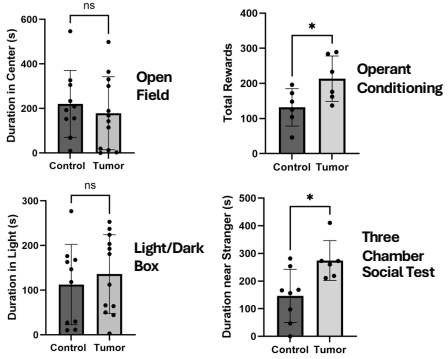


Results

AIM #1: Histology



AIM #2: Behavior

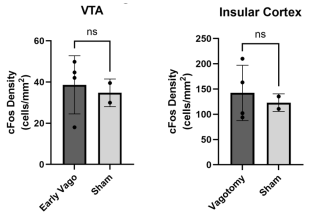


Conclusion

- Early SCLC is associated with changes in brain activity and behavior through tumor–brain communication.
- Significant neural activity changes were initially observed in both the VTA and the IC of male TKO mice with SCLC but were only replicated in the insular cortex.
- Behavioral changes were observed in male TKO mice with SCLC; changes in anxiety-related behaviors were not significant while changes in reward behavior and social interaction were of significance.

Future Directions

- Additional mice are being added for VTA analyses, given the smaller size of this brain region and the lower number of cFos-positive cells.
- Preliminary analyses of vagotomy and sham vagotomy cohorts showed no significant differences in neural activity; additional mice are being added to determine whether vagotomy truly has no effect on activity.



References

Hartmann, G. G., & Sage, J. (2024). Small-cell lung cancer neuronal features and their implications for tumor progression, metastasis, and therapy. Molecular Cancer Research.

Savchuk, S., Gentry, K. M., Wang, W., Carleton, E., Biagi-Junior, C. A. O., Luthria, K., Yalçın, B., Ni, L., Farnsworth, H. C., Davis, R. A., Droxler, R., Meirna, J. C., Liu, Y., Acosta-Alvarez, L., Hartmann, G. G., Pavarino, E. C., LaBelle, J., Woo, P. J., Toland, A. M., & Qu, F. (2025). Neuronal activity-dependent mechanisms of small cell lung cancer pathogenesis. Nature.

Schaffer, B. E., Park, K.-S., Yiu, G., Conklin, J. F., Lin, C., Burkhart, D. L., Karnezis, A. N., E. Alejandro Sweet-Cordero, & Sage, J. (2010). Loss of p130 Accelerates Tumor Development in a Mouse Model for Human Small-Cell Lung Carcinoma. Cancer Research, 70(10), 3877–3883.